



Healthcare Insurance Cost Analysis



Project Overview

This project presents an end-to-end analysis of healthcare insurance costs using **SQL** and **Microsoft Power BI**. The objective is to identify the key factors influencing insurance charges and provide actionable insights through interactive dashboards and data-driven reporting.

The analysis explores how demographic and lifestyle factors such as age, BMI, smoking status, gender, number of children, and region affect insurance costs, enabling insurers to make informed pricing and risk assessment decisions.

This project is part of my Data Analytics portfolio and demonstrates practical SQL querying, data modelling, business intelligence, and dashboard development skills.



Business Problem

Healthcare insurance providers need to understand the factors that drive insurance charges in order to:

- Improve premium pricing strategies
- Assess customer risk accurately
- Identify high-cost customer groups
- Support evidence-based business decisions
- Enhance profitability while maintaining fair pricing

This project uses SQL and Power BI to transform raw insurance data into meaningful business insights.



Dataset

The dataset contains healthcare insurance records with customer demographic, lifestyle, and insurance cost information.

Key Fields

- Age
- Gender
- BMI (Body Mass Index)
- Number of Children
- Smoking Status

- Region
 - Insurance Charges
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✂ Tools & Technologies

- **SQL Server** – Data extraction and querying
 - **Microsoft Power BI** – Data modelling, visualization, and dashboard development
 - **Microsoft Excel** – Initial data inspection and validation
 - **GitHub** – Version control and project documentation
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Project Workflow

1. Data Extraction (SQL)

- Queried the insurance database using SQL
- Selected relevant variables for analysis
- Validated data quality

2. Data Preparation

- Checked for missing values
- Reviewed data consistency
- Prepared the dataset for visualization

3. Data Analysis (Power BI)

Created calculated measures and KPIs to evaluate:

- Average insurance charges
- Gender comparisons
- Regional differences
- Smoking impact
- BMI trends
- Age distribution
- Family size effects

4. Dashboard Development

Built an interactive Power BI dashboard allowing users to filter results by demographic characteristics and explore insurance cost drivers.



Dashboard KPIs

- Total Average Insurance Charge
 - Male Average Charge
 - Female Average Charge
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Analysis Performed

The dashboard provides insights into:

- Average Insurance Cost
 - Insurance Cost by Age
 - Insurance Cost by Gender
 - Insurance Cost by BMI
 - Insurance Cost by Number of Children
 - Insurance Cost by Region
 - Insurance Cost by Smoking Status
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Key Insights

Smoking Status

Smoking has the strongest impact on healthcare insurance costs. Smokers incur significantly higher average charges than non-smokers, reflecting the increased health risks associated with smoking.

Age

Insurance costs generally increase with age, indicating that older policyholders are associated with higher healthcare expenditures.

Body Mass Index (BMI)

Individuals with higher BMI values tend to have higher insurance charges, suggesting that obesity and related health conditions contribute to increased medical costs.

Gender

Average insurance charges are relatively similar between males and females, indicating that gender alone has a limited influence on insurance pricing.

Number of Children

Insurance costs vary across family sizes, although the relationship is less pronounced than age or smoking status.

Regional Analysis

Average insurance charges show only moderate variation across regions, suggesting that lifestyle and individual risk factors have a greater impact than geographic location.



Business Recommendations

- Introduce targeted wellness and smoking cessation programmes to help reduce long-term healthcare costs.
 - Use age and BMI as key factors in actuarial pricing models while ensuring compliance with applicable regulations.
 - Offer preventive healthcare incentives for high-risk customer groups.
 - Develop personalised insurance plans based on customer risk profiles.
 - Continue monitoring regional trends to identify emerging healthcare cost patterns.
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Repository Structure

```
Healthcare-Insurance-Cost-Analysis/  
├── Dataset/  
│   └── healthcare_insurance.csv  
├── SQL/  
│   └── insurance_analysis_queries_screenshots.png  
├── Power BI/  
│   └── Healthcare_Insurance_Cost_Analysis.pbix  
├── Dashboard/  
│   └── dashboard_screenshot.png  
├── Report/  
│   └── Healthcare_Insurance_Report.pdf  
└── README.md
```



Skills Demonstrated

- SQL Query Writing
 - Data Extraction
 - Data Cleaning
 - Data Validation
 - Data Modelling
 - Power BI Dashboard Development
 - DAX Measures
 - KPI Reporting
 - Exploratory Data Analysis (EDA)
 - Business Intelligence
 - Data Visualization
 - Data Storytelling
 - Healthcare Data Analytics
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Dashboard Preview

Include screenshots of:

- Power BI Dashboard
 - SQL Queries
 - KPI Cards
 - Insurance Cost by Smoking Status
 - Insurance Cost by Age
 - Insurance Cost by BMI
 - Insurance Cost by Region
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Future Enhancements

- Develop predictive models to estimate future insurance charges using machine learning.
 - Build customer risk segmentation using clustering techniques.
 - Add forecasting to estimate healthcare costs over time.
 - Integrate additional demographic and medical variables for deeper risk analysis.
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About Me

Dayo Rex Afariogun

Data Analyst with practical experience in **SQL, Microsoft Power BI, Microsoft Excel, and Python**. I enjoy transforming raw data into actionable insights that help organisations make informed business decisions.

Connect with Me

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★ If you found this project useful or interesting, feel free to star this repository or connect with me on LinkedIn.